HDOC Day-49

Finding a Pivot Integer Using Functions and Pointers in C

Aim

To write a C program using a user-defined function and pointers to find a pivot integer x for a given positive integer n. The sum from 1 to x (inclusive) must equal the sum from x to n (inclusive). Display x if found, or -1 otherwise.

Algorithm

1. Start the program and declare n and result.
2. Read a positive integer n. Reject values less than or equal to zero.
3. Pass the address of n to pivot_integer() using &n.
4. Inside pivot_integer(), access the value using *n.
5. For each candidate x from 1 through n, calculate the sum of 1 through x and the sum of x through n.
6. If the sums match, return x; otherwise, continue searching.
7. Return -1 if no pivot exists.
8. Display the result in main() and stop.

C Program

```c
#include <stdio.h>

int pivot_integer(int *n);

int main()
{
    int n, result;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid input! Enter a positive integer.\n");
        return 0;
    }

    result = pivot_integer(&n);

    if (result != -1)
        printf("Pivot Integer: %d\n", result);
    else
    {
        printf("No pivot integer exists.\n");
        printf("Result: -1\n");
    }

    return 0;
}

/* Function to find the pivot integer */
int pivot_integer(int *n)
{
    int x, i;
    long long leftSum, rightSum;

    for (x = 1; x <= *n; x++)
    {
        leftSum = 0;
        rightSum = 0;

        /* Sum of integers from 1 to x */
        for (i = 1; i <= x; i++)
            leftSum += i;

        /* Sum of integers from x to n */
        for (i = x; i <= *n; i++)
            rightSum += i;

        if (leftSum == rightSum)
            return x;
    }

    return -1;
}
```

Explanation

The main function passes &n to pivot_integer(), which receives an int pointer. The expression *n reads the value at that address. For each x, the function compares the inclusive sums 1 + 2 + ... + x and x + (x + 1) + ... + n.

Test Case Table

Case	Input n	Expected Result
1	8	Pivot Integer = 6
2	1	Pivot Integer = 1
3	4	-1
4	3	-1
5	49	Pivot Integer = 35
6	2	-1
7	0	Invalid input
8	-5	Invalid input

Sample Outputs

Pivot integer exists

```
Enter a positive integer: 8
Pivot Integer: 6
```

Pivot integer does not exist

```
Enter a positive integer: 4
No pivot integer exists.
Result: -1
```

Single positive integer

```
Enter a positive integer: 1
Pivot Integer: 1
```

Invalid input

```
Enter a positive integer: -5
Invalid input! Enter a positive integer.
```

Verification for n = 8: 1 + 2 + 3 + 4 + 5 + 6 = 21 and 6 + 7 + 8 = 21, so the pivot integer is 6.

Result

The C program was designed using a user-defined function, pointers, loops, and conditional statements. The pivot_integer() function receives the address of n and returns the pivot integer when the two inclusive sums are equal, or -1 if none exists.